Gosford High School
Year 9 Computing Technology
Assessment Task 2026

Number and Title of Task	Assessment Task 2: In-Class Project (35%)
Topic	Building Mechatronic and Automated Systems
Weighting/Marks	35% (70 marks)
Due Date	Term 3 Week 5, Friday 21st August 2026

Outcomes to be Assessed (Syllabus outcomes being assessed by task)

A student:

CT5-DPM-01 applies iterative processes to define problems and plan, design, develop and evaluate computing solutions
CT5-COL-01 manages, documents and explains individual and collaborative work practices
CT5-EVL-01 understands how innovation, enterprise and automation have inspired the evolution of computing technology
CT5-OPL-01 designs, produces and evaluates algorithms and implements them in a general-purpose and/or object-oriented programming language
CT5-THI-01 applies computational, design and systems thinking to the development of computing solutions

Introduction

In this project, you will work individually to design, program, and test a Raspberry Pi PICO microcontroller device to solve a specific problem or address a particular need. You will be provided with a kit and will need to use **at least one additional sensor** from an external sensor kit as part of your design **for full marks**.

Part 1: Project Documentation (40 marks)

- **Requirements Definition:** Determine a list of functional and non-functional requirements that your robot must meet to address the problem effectively.
- **Design:** Design a pseudocode algorithm for a mainline routine indicating how the overall program may run and two pseudocode algorithms for subroutines outlining how some of your functions will work. You will also develop a flowchart that visually represents each algorithm.
- **Peer Evaluation:** You and your classmates will complete peer evaluations of each other's projects upon completion and evaluate your project in relation to this peer feedback.
- **Project Evaluation:** You will evaluate your project in relation to functional and non-functional success criteria, performance, project management and provide suggestions for software and hardware improvements to enhance the mechatronic solution.
- **Documentation:** You will document your project's progress in a Markdown file and make regular comments in your GitHub commits. This documentation will serve as a record of your project's development.

Part 2: Testing and Implementing (30 marks)

- **Testing and Debugging:** For each test, you will copy your code into your Markdown file and evaluate test cases, along with debugging processes. Clear evidence of testing and debugging will be essential.
- **Code and Comments:** All individual test code needs to be well-structured and made more efficient with features such as control structures (e.g. processes, branching, loops), data structures (e.g. lists, dictionaries) and subroutines (functions). All code must be explained comprehensively with comments (single line) and docstrings (multiline).
- **Final Product:** You need to film your final product functioning and submit this to Google Classroom.

Additional Information (specific timing, additional resources, criteria, etc)

- You will be provided with all required resources, including microcontrollers and wiring, in class.

- If you require additional time to study for the test or complete your project's documentation, you may need to do some of this for homework, or attend Hack Club on Monday during lunch times to seek peer support. **No additional Recess or Lunch Times will be granted beyond Hack Club.**

- If you are unsure of any element of the task, it is **YOUR** responsibility to seek advice and support in a timely manner. **Do NOT wait until the week the task is due to seek clarification!**

Feedback (How you will receive feedback and reflect on the task)

Teacher feedback will include:

- Verbal formative feedback on project development
- Individual written summative feedback
- Verbal summative feedback as a class group

Students will reflect on their task through:

- Completing detailed documentation of their processes
- Peer evaluation
- Evaluating their project's success, performance, and necessary improvements

Student Acknowledgement of Academic Integrity

By submitting the task for marking, I acknowledge the following:

1. The work submitted is my own work and appropriately acknowledges of all sources has been made.
2. I have not used generative AI in the **construction** of the task.
3. If there is an allegation of malpractice, I will be required to show my drafts/ planning to prove the task is all my own work.
4. I am aware that the work may be submitted to plagiarism detection processes.
5. Where the work of others is used and not acknowledged, a finding of plagiarism will be made and a mark of zero awarded and I will have to resubmit the task.

Part A: Project Documentation (30 marks)

/30	5 Marks (Per row)	4 Marks (Per row)	3 Marks (Per row)	2 Marks (Per row)	1-0 Marks (Per row)
Requirements Outline **CT5-DPM-01**	Comprehensive explanation of functional requirements, non-functional requirements and test cases.	Thorough description of functional requirements, non-functional requirements and test cases.	Expected outline of functional requirements, non-functional requirements and test cases, but needed more description.	Basic identification of functional requirements, non-functional requirements and/or test cases are identified, may lack details.	Incomplete or missing functional requirements, non-functional requirements and/or test cases.
Pseudocode Development **CT5-OPL-01**	3 pseudocode algorithms are substantially correct, well-structured, and highly detailed.	3 pseudocode algorithms are mostly correct and reasonably structured.	3 pseudocode algorithms are provided with some errors or omissions.	At least 2 pseudocode algorithms are provided but are largely incorrect or lacks detail.	No pseudocode provided or pseudocode is limited with minimal structure.
Flowchart Development **CT5-OPL-01**	3 flowcharts are substantially correct and perfectly represent the pseudocode with no inaccuracies.	3 flowcharts represent the pseudocode with minor inaccuracies.	3 flowcharts represent the pseudocode with some inaccuracies.	At least 2 flowcharts are provided but do not reflect the pseudocode.	No flowchart provided or flowchart does not represent the pseudocode.
Peer Feedback and Reflection **CT5-COL-01**	Sophisticated evaluation of peer feedback demonstrating insightful reflection and understanding of areas for improvement.	Thorough discussion of peer feedback demonstrating a good understanding of areas for improvement.	Expected explanation of peer feedback demonstrating a reasonable understanding of areas for improvement.	Basic description of peer feedback, but may lack willingness to consider potential areas for improvement.	Limited or no consideration of peer feedback.
Project Evaluation **CT5-OPL-01, CT5-DPM-01**	Sophisticated project evaluation including detailed assessment of functional and non-functional criteria, performance, project management and actionable suggestions for improvement.	Detailed project evaluation including thorough discussion of functional and non-functional criteria, performance, project management and suggestions for improvement.	Expected project evaluation including reasonable explanation of functional and non-functional criteria, performance, project management, but may not fully address areas for improvement.	Basic project evaluation including some description functional and non-functional criteria, performance, project management and/or suggestions for improvement.	Limited project evaluation which does not fully address criteria, performance, project management or suggestions for improvement.
Version Control **CT5-DPM-01 CT5-COL-01**	Extensive and consistent use of GitHub to document development. Commits are meaningful, frequent, and clearly describe progress and iterations.	Proficient and consistent use of GitHub to document development. Most commits are relevant and track progress adequately.	Expected use of GitHub commits are made to document development. However, commit messages may lack clarity or consistency.	Progressing use of GitHub. Few commits are made and/or commit messages are often unclear or unrelated to specific stages of development.	Little to no meaningful use of GitHub. Commits are absent, minimal, or do not reflect actual development progress.

Part B: Testing and Implementation (40 marks)

/20	20 Marks (Per row)	16 Marks (Per row)	12 Marks (Per row)	8 Marks (Per row)	4-0 Marks (Per row)
Code and Comments **CT5-OPL-01**	Highly sophisticated program with code that includes: - Extensive level of functionality - Outstanding use of control structures and data structures for optimisation and efficiency - Clear comments and docstrings which acknowledge external sources such as websites or AI - Any AI support or external code use is acknowledged and only for very specific purposes	Advanced program with code that includes: - High level of functionality - Advanced use of control structures and data structures for optimisation and efficiency - Mostly clear comments and docstrings which acknowledge external sources such as websites or AI - Any AI support or external code use is acknowledged and only for very specific purposes	Expected program with code that includes: - Expected level of functionality - Expected use of control structures and/or data structures for optimisation and efficiency, but may overuse IF statements / lack subroutines - Somewhat clear comments and docstrings which acknowledge external sources such as websites or AI - Any AI support or external code use is acknowledged and only for specific purposes	Basic program which may have been created with assistance (e.g. AI, external sources), with code that includes: - Basic level of functionality, may include errors - Progressing use of control structures and/or data structures, may lack optimisation or efficiency due to overuse of IF statements and repetitive code - Basic comments and/or docstrings which do not properly acknowledge external sources such as websites or AI - AI support or external code use is either unacknowledged or too heavily relied upon to create the program.	Limited programming techniques utilised with assistance (e.g. AI, external sources), with code that includes: - Limited level of functionality, may include significant errors - Limited use of control structures and/or data structures - Limited or no comments or docstrings which do not properly acknowledge external sources such as websites or AI - AI support or external code may have been used to create most of the program
/20	**10 Marks (Per row)**	**8 Marks (Per row)**	**6 Marks (Per row)**	**4 Marks (Per row)**	**2-0 Marks (Per row)**
Testing and Debugging **CT5-THI-01, CT5-OPL-01**	Comprehensive evidence of extensive testing and effective debugging processes, with several code blocks and sophisticated evaluation.	Detailed evidence of testing and effective debugging processes, with several code blocks and thorough evaluation.	Sufficient evidence of testing and debugging processes, with some code blocks and satisfactory explanation.	Basic evidence of testing and debugging processes, with one or two code blocks and elementary description.	Limited or no evidence of testing and debugging processes.
Final Product **CT5-OPL-01, CT5-COL-01**	Excellent final product which exceeds expectations in terms of initial need, requirements and test cases and makes outstanding use of additional sensor(s).	High quality final product which meets expectations in terms of initial need, requirements and test cases and makes proficient use of an additional sensor.	Expected final product which meets most expectations in terms of initial need, requirements and test cases, but may not include an additional sensor and may have small flaws and/or inconsistent performance.	Basic final product which meets few expectations in terms of initial need, requirements and/or test cases, does not utilise an additional sensor and may have significant performance issues.	Limited or no final product which does not meet expectations in terms of initial need, requirements, test requirements and does not include an additional sensor.

Total /70

Feedback: